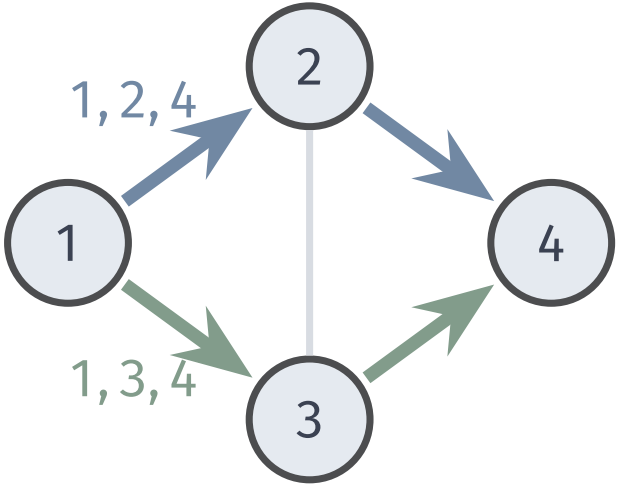

(a)

	1	2	3	4
1	2	1	1	2
2	1	3	2	1
3	1	2	3	1
4	2	1	1	2

(b)

$$(\mathbf{A}^2)_{1,4} = \sum_\ell a_{1,\ell}\, a_{\ell,4}$$

$$= a_{1,2}a_{2,4} + a_{1,3}a_{3,4}$$

$$= 1 + 1 = 2$$